- **Colour** RBC appears yellow when seen singly, but these appear red when in bulk due to the haemoglobin.
- **Formation** In the developing foetus, the formation of RBC takes place in liver and spleen, while after the birth, RBCs are mainly produced in the bone marrow.
- Maturation of RBC is controlled by folic acid and vitamin-B_{12}. The excess of RBCs are stored in the spleen, which acts as a blood bank of the body.

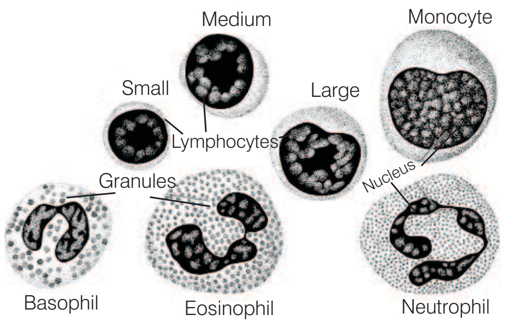
Functions of RBCs

- Haemoglobin helps in maintaining a constant pH.
- Haemoglobin also transports carbon dioxide from the tissues to the lungs and oxygen from the lungs to body tissues.

White Blood Corpuscles (WBCs)

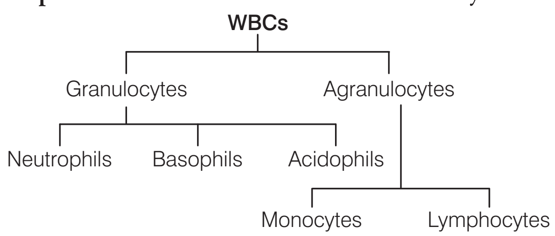
WBCs are also called **leucocytes** or **leukocytes**. These are the cells, which are important part of immune system and involved in protecting the body against both infectious disease and foreign invaders.

- **Shape** These are rounded or amoeboid (irregular), nucleated, non-pigmented cells. WBCs are larger in size than RBCs.
- **Number** WBCs are much less in number than RBCs (1:6).



Structure of WBC

- **Leukemia** (blood cancer) is a pathological increase in number of WBCs. Fall in WBC count is termed as **leucopenia**.
- **Colour** WBCs are colourless.
- **Formation** WBCs are formed in red bone marrow.
- **Lifespan** WBCs survive for 8-96 hours only.



Classification of WBCs

Granulocytes These are about 65% of total leucocytes. These are subdivided on the basis of shapes of their nuclei and the staining reactions of their granules.

(*i*) **Neutrophils**
- These are about 62% of total number of WBCs.
- They originate from bone marrow.
- The cytoplasm of these contains fine granules.
- Their nucleus is 3-5 lobed and lifespan is about 10-12 hours.
- These also help in sex differentation and act as soldiers of body.

(*ii*) **Basophils**
- These are also called **cyanophils**. Their nucleus is 2 or 3-lobed or S-shaped and lifespan is about 8-12 hours.
- These secrete heparin and histamine, thus help in local anticoagulation.

(*iii*) **Acidophils**
- These are also called **eosinophils**.
- Their lifespan is about 14 hours.
- These increase in number in allergic diseases.
- These help in healing of wounds.

Agranulocytes Formation of agranulocytes is also called **agranulopoiesis**. These form about 35% of the total WBCs. These are divided in two subtypes:

(*i*) **Monocytes**
- These are the largest sized leucocytes and form about 5.3% of all the leucocytes. The nucleus is oval, kidney or horse shoe-shaped and is excentric.
- These are formed in the lymph nodes and the spleen.
- These are highly motile and their lifespan is of 3-4 days.

(*ii*) **Lymphocytes**
- These form about 30% of leucocytes.
- Their nucleus is large and rounded, so that cytoplasm forms a thin peripheral layer. They are formed in bone marrow.
- These are formed in the thymus, lymph nodes, spleen, tonsils, etc.
- Their lifespan is of 3-4 days.
- These produce antibodies and opsonin, so play an important role in immune system.

Blood Platelets

- **Shape** These are oval-shaped and possess discoidal cytoplasmic fragments.
- **Number** Their number varies from 250000-5000000. A decrease in the number of platelets in the blood is called **thrombocytopenia**. Increase in number of blood platelets is called **thrombocytosis**.
- **Structure** Platelets are flat, non-nucleated large cells in the bone marrow. They are bounded by a membrane and contain few organelles in cytoplasm. These play an important role in blood clotting.
- **Colour** They are colourless, formed in red bone marrow.
- **Lifespan** about one week.

Blood Groups

Landsteiner recognised three kinds of blood groups 'A', 'B' and 'O'. Fourth and most rare 'AB' blood group was discovered by **Von Decastello** and **Sturle** (1902).

Landsteiner (1900) discovered two kinds of antigens called a and b. Antigens a and b are not proteins, but are mucopolysaccharides.

Characteristics of Human Blood Groups

Blood group	*Antigens in RBCs*	*Antibodies in plasma*	*Can donate to groups*	*Can receive from groups*
A	a	Anti-b	A, AB	O, A
B	b	Anti-a	B, AB	O, B
AB	a and b	None	AB	O, A, B, AB
O	None	Anti-a Anti-b	O, A, B, AB	O

- Persons with 'O' blood group are called as **universal donors** and AB are **universal recepient**.

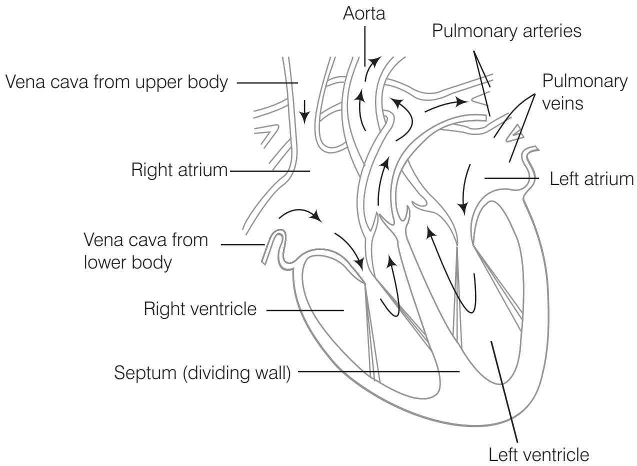
Rh-Factor

Rh-factor was first of all reported in RBCs of a monkey by **Landsteiner** and **Wiener** in 1940.

- Rh-factor is actually the Rh antigen present on the surface of red blood cells.
- Person having Rh-factor is called $Rh^+$and without Rh factor is Rh^-.
- For an example, if person of blood group A has Rh antigen on surface of RBC, then he is A^+/A positive.
- About 85% of world's people are Rh^+.
- Percentage of Rh^+ people in India is 97% of the population.
- Rh-factor is found in man and rhesus monkey, it has not been reported from other animals.

3. Heart

It is a muscular organ in both humans and other animals, which help in pumping blood *via* blood vessels of the circulatory system.



Internal Structure of Heart

- In amphibians, heart is three-chambered.
- Reptilian heart is structurally three-chambered, but is functionally four-chambered (i.e. incompletely four-chambered) except in crocodile.
- In crocodile, birds and mammals the heart is divided into four chambers (two auricles and two ventricles).
- Every person's heart is of about the same size as that of person's first. The human heart is four-chambered, two auricles and two ventricles.
- A thin, muscular wall called the interatrial septum separates the left and right atria, whereas thick-walled interventricular septum separates the right and left ventricles.
- The atrium and ventricle of same side are separated by atrioventricular septum.
- Opening between right atrium and right ventricle is tricuspid valve, formed of 3 cusps and opening between left atrium and left ventricle is called **bicuspid** or **mitral** valve.
- Pulmonary artery and aorta are provided with semilunar valves to prevent backflow of blood.
- Sino Atrial Node (SAN) is present in right upper corner of right atrium. Atrio Ventricular Node (AVN) is present in lower left corner of right atrium.

Cardiac Cycle

The sequence of events, which occur from the beginning of one heartbeat to the beginning of next is called **cardiac cycle**. Systole refers to contraction and diastole refers to relaxation. Single cardiac cycle involves four steps:

1. **Atrial Systole**

- Right atrium receives deoxygenated blood from superior and inferior vena cava.
- Left atrium receives oxygenated blood through pulmonary veins.
- The atria contract due to wave of contraction stimulated by SA node.

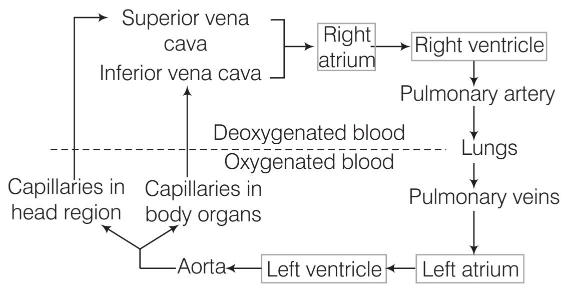
2. **Ventricular Systole**

- Action potential passes from SA node to AV node. The blood is then forced into ventricles leading to contraction of ventricles.
- At this time first heart sound is produced due to closure of valves.
- When the contraction is complete blood flows to pulmonary arteries carrying deoxygenated blood to lungs.

3. **Ventricular Diastole**

- Semilunar valves close to produce second heart sound.
- The ventricular relax. AV valve opens to allow blood to pass from atria to ventricles again.

Left ventricles supply oxygenated blood to aorta, which in turn supplies blood to various body parts.



Circulation of blood through the whole body

Blood Pressure (BP)

The pressure exerted by the blood on the walls of the blood vessels due to the repeated pumping of heart is called blood pressure. The rate of pulsation increases during excitement.

- Blood pressure is recorded as systolic/diastolic.
- Blood pressure in a normal adult person = 120/80 mm Hg.
- If a person has persistent high blood pressure then it is called **hypertension**. To avoid hypertension good cholesterol, i.e. high density lipoprotein should be consumed.

IMPORTANT POINTS TO REMEMBER

- Haemoglobin can readily combine with carbon monoxide.
- Blood does not coagulate inside the body due anti-coagulant heparin.
- The crew and passanger of aircraft usually suffer from pulmonary problems because of the presence nitrogen oxide in the atmosphere.
- Coronary arteries supply blood to the right atrium.
- Hepatic portal system and renal portal system supply and take blood from liver and kidney respectively.

- Hypotension is condition of low blood pressure.
- **ECG** or Electrocardiogram is a graphic record of the electric current produced by the excitation of cardiac muscles.
- The normal rate of heartbeat at rest is about 72-80 beats per minute.
- In a newly born baby, heartbeat rate is about 140 beats per minute.
- During heavy exercises, it may be high as 170-200 beats per minute.
- **Sphygmomanometer** measures BP.

EXCRETORY SYSTEM

The process of removal of wastes from the body is called **excretion**. The organs of excretion are called **excretory organs**.

Excretory Organs in Different Animals

- **Plasmalemma** Protozoans like *Amoeba*.
- **General body surface** Porifera (sponges), coelenterates (*Hydra*).
- **Flame cells** Platyhelminthes (*Taenia* and *Fasciola*).
- **Nephridia** Annelida.
- **Malpighian tubules** Arthropods (cockroach).
- **Coxal gland** Spiders.
- **Kidney** Main excretory organ in all vertebrates.
- **Antennal gland** or **green gland** Crustaceans crayfish.

Types of Excretion

Depending upon the form of nitrogenous waste excreted, there are three modes of excretion:

1. Ammonotelism

- It is the elimination of nitrogenous end product (waste) mainly as ammonia.
- Ammonotelic excretion is found in aquatic animals like protozoans, e.g. *Amoeba*, *Paramecium*, sponges (e.g. *Sycon*), coelenterates (e.g. *Hydra*), aquatic arthropods (e.g. *Prawn*), most aquatic molluscs (e.g. *Pila*), bony fishes (e.g. *Labeo*) and frog's tadpole.
- Those organisms, which secrete ammonia are called **ammonotelic animals**.

2. Ureotelism

- It is the process, in which main nitrogenous waste is urea.
- It is a common method of excretion in man, whales, seals, camels, kangaroo, toads, frogs, sharks, etc.

3. Uricotelism

- Main nitrogenous waste is crystals of uric acid.
- It is common in birds, land reptiles (snakes and lizards), insects, snails, etc.

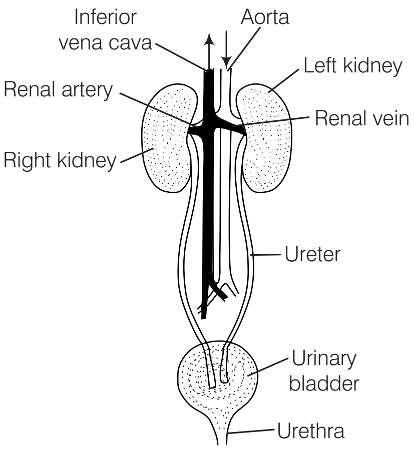
Excretory System of Human

Excretory system of human consists of the following parts:

(a) Kidneys (two)
(b) Ureter (two)
(c) Urinary bladder (one)
(d) Urethra (one)

1. Kidneys

- It is a bean-shaped, chocolate brown structure lying in the abdomen, one on each side of the vertebral column just below the diaphragm.
- These form the urine and control osmotic pressure within the organism with respect to its external environment.
- The left kidney is placed a little higher than the right kidney (but reverse in rabbit).
- Concavity of kidney called **hilum** or **hilus**, is always inward directed.



Structure of human excretory system

2. Ureters

- These are a pair of narrow tube arising from the hilum and is called as **ureter**.
- It is about 30 cm in length.
- They bring the urine downwards and open into urinary bladder.

3. Urinary Bladder

- It is a medium pear-shaped sac situated in the lower or pelvic region of the abdominal cavity within the body wall.
- Each ureter opens in urinary bladder.
- It temporarily stores urine.
- It is absent in birds. In both reptiles and birds, ureters and rectum open into a common sac called **cloaca**.

4. Urethra

- It is a muscular and tubular structure, which extends from neck of bladder to outside.
- In females, this tube is small and serves as a passage of urine only, while in males, it is long and functions as a common passage for urine and spermatic fluids.

Nephron

- It is the structural and functional unit of kidney.
- These are also called as **renal tubules** or **uriniferous tubules**.
- It is differentiated into four regions having different anatomical features and physiological role.
  (*i*) Bowman's capsule.
  (*ii*) Proximal Convoluted Tubule (PCT).
  (*iii*) Henle's loop U-shaped.
  (*iv*) Distal Convoluted Tubule (DCT).

Bowman's Capsule

- It is a double-walled cup and is lined by thin flat cells called **podocytes**.
- It contains group of capillaries called **glomerulus**.
- Glomerulus in the kidney acts as a dialysing bag.

Proximal Convoluted Tubule (PCT)

- It is a highly coiled (convoluted) tubular structure.
- It is about 12-24 mm in length.
- Almost whole of vitamins, glucose, amino acids, sodium and potassium, etc., is reabsorbed here, mainly by active transport.

Henle's Loop

- It is a U-shaped segment of nephron located in renal medulla.
- Loop of Henle is long in mammals and birds, which secrete hypertonic urine, but short in other vertebrates like reptiles, etc.

Distal Convoluted Tubule (DCT)

- It is similar to PCT region, i.e. greatly curved twisted.
- It is connected to the collecting duct.
- Active reabsorption of some Na^+ takes place here.
- It is impermeable to H_2O.
- Renal tubules along with large intestine accounts for most of the absorption of water in body.

Urine

Urine is formed after glomerular filtration, reabsorption and secretion. It is pale-yellow coloured transparent fluid due to the presence of urochrome pigment. It is acidic in nature and is slightly heavier than water.

- It has a faint aromatic odour due to urinod.
- Daily urine output in normal adult is 1.5-1.8 L.
- Chemical composition of urine, i.e. water is 95-96%, urea is 2% and some other substances like uric acid, creatinine, etc., are 2-3%.

Skeleton System

The human skeleton consists of both fused and individual bones supported and supplemented by ligaments, tendons, muscles and cartilage, ligaments and tendons are connective tissue made up of collagen fibres.

It is divided into two parts:

(i) Axial Skeleton (80 Bones)

- It includes skull, vertebral column and bones of chest.
- Vertebral column is responsible for the upright position of the human body.
- Most of the body weight is located in back of the spinal column.
- It provides flexibility to the neck and protection to spinal cord.
- The total number of bones of head is 29.
- The total number of bones in vertebral column is initially
  33 and after development is 26.
- The total number of bones of ribs is 24.

(ii) Appendicular Skeleton (126 Bones)

- It consists of pelvic and pectoral girdles.
- Girdles are the structures, which connect the limb bones with the vertebral column.
- Their functions are to make locomotion possible and to protect the major organs of locomotion, digestion, excretion and reproduction.
- Calcium and phosphorous are two minerals that helps in giving strength to bones. Deficiency of these may cause we opening of bones.

NERVOUS SYSTEM

The nervous system provides the fastest means of communication within the body, so that suitable response to stimuli can be made at once.

Nervous system is found only in animals and absent in plants. It is made up of nervous tissue. Nervous system controls the body by using a series of tissues throughout the body formed by a network of electrically conducting cells called **neurons** or **nerve cells**.

Neuron

It is the structural and functional unit of nervous system.

Neurons are of three types:

(*i*) **Motor** Conducts messages from central nervous system to effector organs.

(*ii*) **Sensory** Conducts information from sensory organ to central nervous system.

(*iii*) **Mixed** Works both as sensory and motor neuron.

Structure of Neuron

A neuron is a microscopic structure composed of three major parts, i.e. cell body, dendrites and axon.

- **Dendrites** Cytoplasmic processes from cell body. They carry impulses towards cell body.
- **Cell body** Also called **cyton** or **soma**. It contains cytoplasm and certain granular bodies called **Nissl's granules**.
- **Axon** Single long extension of cell body that protected by membrane neurolemma.
- **Synapse** It is the junction between the dendrites of one neuron with axon terminal knobs of other neuron.

Classification of Neuron/Nerve Cell

Neurons are of two types:

1. **Myelinated** Axon is composed of Schwann cell coated by myelin sheath. The gap between adjacent myelin sheath is called **nodes of Ranvier**.

On the basis of number of axon and dendrites
- Unipolar (with one axon)
- Bipolar (with one axon and one dendrite)
- Multipolar (with one axon and two or more dendrites)

2. **Non-myelinated** Axon is composed of **Schwann** cells not enclosed by myelin sheath.

Types of Nervous System

Main parts of nervous system are as follow:

Main parts of vertebrate nervous system
- Central nervous system
  - Brain
  - Spinal cord
- Peripheral nervous system
  - Somatic nervous system
  - Autonomic nervous system
    - Parasympathetic nervous system
    - Sympathetic nervous system

Central Nervous System (CNS)

The CNS is the site of information processing and control. It consists of brain and spinal cord.

- Grey matter is made up of cell bodies of neuron and white matter has only nerve fibres.
- Brain lies in the cranial cavity of skull. Brain and spinal cord are covered by two meninges in frog and three meninges in mammals.
- Cerebrospinal fluid is present in and around brain and spinal cord.

Human Brain

Brain is soft, whitish in colour and somewhat a flattened organ.

- It is the only thing in the universe, which is attempting to understand itself.
- It is mainly divided into three main parts:

Brain
- Forebrain
  - Cerebrum
  - Olfactory lobes
  - Thalamus (Hypothalamus)
- Midbrain
  - Corpora quadrigemina
  - Corpora diencephalon
- Hindbrain
  - Cerebellum
  - Medulla oblongata
  - Pons

Forebrain

It forms the greatest part of brain. It consists of following types:

- **Cerebrum** It is the largest part of the brain covered by cerebral hemispheres. The roof of cerebrum is cerebral cortex. Cerebrum leads to consciousness, storage of memory of information.
- **Olfactory lobes** These are pair of small sized structure, completely covered by cerebrum. These are the centre of skull.
- **Thalamus** It is for integrity of sensory impulses sent from sense organs like eyes, ears and skin. It deals with pain, pressure and temperature.